

Exhaustive Boundary-Layer Certification for Megaminx via Neural-Guided Beam Cascades

Vladislav Kuznetsov

Moscow Institute of Physics and Technology (MIPT), Dolgoprudny, Russia

July 2026

Abstract

We certify one-move reductions for two adjacent phase pairs in a subgroup-chain Megaminx solver under the face-turn metric. The separate worst-case solving distances of phases 3 and 4 are 14 and 8, while those of phases 5 and 6 are 13 and 13. A boundary-layer argument shows that improving either sum by one requires checking only compositions for which both constituent coordinates are maximally distant from their target subgroups. This reduces combined quotients of sizes 14,234,011,545,600 and 1,664,617,163,489,280 to exhaustive frontiers of 536,572 and 407,628 physical states. Exact local rewrites certify 8,664 states. For the remaining 935,536, compact neural action-ranking models guide batched beam-search cascades that produce bounded solution words. The models and search scores are not proof objects. A deterministic checker requires exact frontier coverage, while separately implemented replay engines validate every word proposed by neural-guided search; exact-rewrite words are also replayed physically. In the published run the models supplied 935,536 direct certificates in approximately 6.25 active A100 GPU-hours, excluding training, preprocessing, and verification. We thereby obtain the ambient-FTM bounds

$$\mathcal{R}_\Omega(G_5, G_7) \leq 21, \quad \mathcal{R}_\Omega(G_7, G_9) \leq 25.$$

Relative to the externally reported 114-move phase chain, these local results give the conditional bound 112; together with the separately certified color-neutral first-phase result, they give the likewise conditional bound 111. The paper is accompanied by the complete certificate databases, pinned phase-table reconstruction, and fail-closed verification software.

1 Introduction

The Megaminx is a face-turning permutation puzzle with 12 pentagonal faces, 20 movable corners, and 30 movable edges. Its state space is too large for a direct breadth-first computation of the maximum solution length. As in the largest certified computations for the Rubik's Cube, useful upper bounds are therefore obtained by decomposing the solve along a subgroup chain and certifying the worst-case cost of each transition [13, 14, 19].

The phase chain studied here was developed in the public Megaminx solver of Botz [5]. In the face-turn metric, its separately computed phase bounds include the consecutive pairs

$$14 + 8 = 22 \quad \text{and} \quad 13 + 13 = 26.$$

The complete externally reported chain has length 114. This paper does not reconstruct the computations outside the two phase pairs considered below; consequences for that global value are therefore stated conditionally.

Our starting point is a simple boundary-layer observation. To lower a sum $a + b$ of two integer-valued phase bounds to $a + b - 1$, every state for which at least one constituent phase has distance below its maximum is already harmless. Only states whose two coordinates are simultaneously maximal require new solutions. For phases 3 and 4 this boundary consists of physical lifts of the Cartesian product of 212 and 2,531 coordinates; for phases 5 and 6 it is a product of 3,484 and 117 coordinates. Thus two combined quotients containing roughly 1.42×10^{13} and 1.66×10^{15} coordinates reduce to 536,572 and 407,628 explicit proof obligations.

Closing nearly one million individually specified obligations remains a substantial search problem. We train compact neural models directly on the two combined quotients using sparse supervision from random trajectories. The models rank actions inside one beam per root, evaluated in batches across many roots on a GPU. A staged schedule applies narrow beams to the full unresolved set and reserves wider beams for the diminishing survivor tail. Exact local rewriting closes 8,664 states, and the neural-guided search produces bounded words for the remaining 935,536 in approximately 6.25 active A100 GPU-hours. This aggregate excludes model training, phase tables, exact rewriting, process gaps, and final replay.

The neural computation is deliberately outside the trusted proof base. A checkpoint need not estimate a true distance, and an individual beam search need not be complete. The proof uses only the resulting words: the checker requires an exact partition of the exhaustive frontier, binds every record to its physical state, enforces the pair-specific length limit, and replays every direct word without loading a model. This yields the certified ambient-FTM upper bounds

$$\mathcal{R}_\Omega(G_5, G_7) \leq 21, \quad \mathcal{R}_\Omega(G_7, G_9) \leq 25.$$

The contributions are therefore:

1. a general boundary-layer reduction for improving two consecutive subgroup phases by one move;
2. complete physical constructions of the two simultaneous-maximal frontiers, including exact lifting and coordinate round trips;
3. a proof-oriented neural beam cascade that generates independently checkable words for a fixed exhaustive worst-case frontier rather than a sampled test distribution; and
4. deterministic certificate checks establishing the two local bounds and, conditional on the reported surrounding phase chain, the global reductions $114 \rightarrow 112$ and $114 \rightarrow 111$ when combined with the separate color-neutral first-phase result.

We do not claim a new generic beam-search algorithm, exact combined distances, or ordinary graph diameters. In particular, combined paths are allowed to leave their source subgroups under the full FTM move alphabet. The mathematical result is exhaustive coverage by bounded words; the learned search is an effective way of discovering those words.

2 Related work

Certified bounds for permutation puzzles. Group actions, Cayley graphs, and subgroup methods provide the standard language for permutation puzzles [12]. Pattern databases and memory-efficient heuristic search made large puzzle spaces computationally accessible [13, 9]. Kunkle and Cooperman used computational group methods to establish a 26-move upper bound for the Rubik’s Cube [14]; Rokicki, Kociemba, Davidson, and Dethridge later certified that the diameter in the half-turn metric is 20 [19]. These works exemplify the separation between a large discovery computation and a smaller, auditable verification argument. The present computation uses the same high-level discipline, but its finite obligation is a pair of maximal-layer products in a Megaminx subgroup chain.

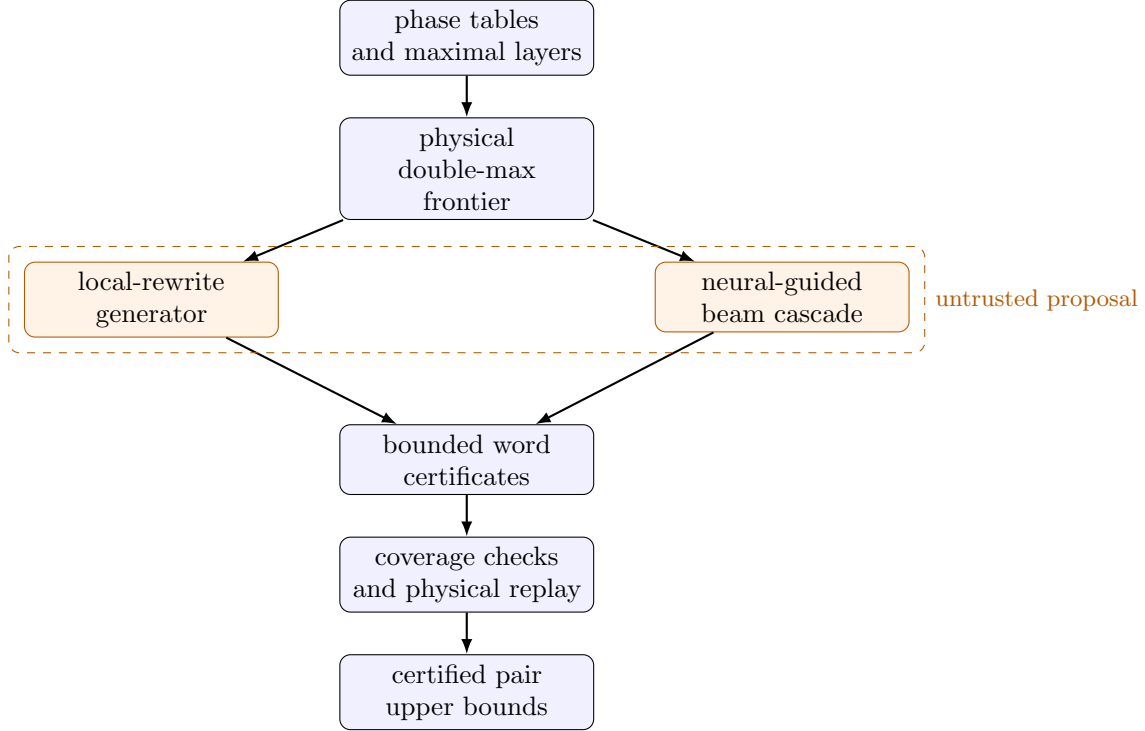


Figure 1: Proof pipeline and trust boundary. Both search procedures propose words; neither procedure is trusted. The proof consumes only the exhaustive frontier, the stored words, and the results of deterministic coverage and replay checks.

Learned heuristic search. DeepCubeA combines a learned cost-to-go function with batched heuristic search and solves random instances of several puzzles, including the Rubik’s Cube [1]. EfficientCube demonstrates that comparatively simple self-supervision from goal-rooted scrambles can support neural-guided beam search [21]. More recent work uses neural distance models and massively parallel beam search on larger Rubik’s Cubes [7]. Our training signal and model architecture are secondary to the certificate result: the models need only rank sufficiently many promising actions to find a valid word for every member of a fixed exhaustive frontier.

Beam portfolios and batched learned heuristics. Restarting a learned beam search with progressively larger widths has appeared in combinatorial optimization [8]. GPU batching of expensive learned heuristics is likewise established in heuristic-search research [11, 10]. Accordingly, we describe our procedure as a *beam cascade* or staged search portfolio, not as a new beam-search algorithm. Its role in our computation is proof-oriented budget allocation: each stage searches only unresolved members of an already exhaustive family, and each success is persisted as a model-independent certificate.

Megaminx computations. The coordinates, phase numbering, move conventions, and reported global bound used here originate in the software and accompanying analysis of Botz [5, 2]. The public history begins with the 116-move chain [3]. A later table correction confirmed, rather than improved, that value [4]; two subsequent reports reduced the last-phase contribution $26 \rightarrow 25 \rightarrow 24$, giving $116 \rightarrow 115 \rightarrow 114$ [17, 18, 6]. These are external software and forum reports, not computations regenerated in this paper. The separate phase-1 result proves that allowing the first block target to vary over all 20 corners lowers its worst-case cost

from 10 to 9 [15]. Neither that result nor the present paper reruns all external computations behind the reported 114-move chain; the local theorems are unconditional, whereas the global arithmetic is not.

3 Metric, coordinates, and boundary layers

3.1 Face-turn metric and subgroup coordinates

Let a face symbol denote a clockwise turn through one fifth of a revolution. For a set F of faces, write

$$\Sigma_F = \{X^k : X \in F, 1 \leq k \leq 4\}.$$

Every element of Σ_F has cost one. Thus X , X^2 , X^3 , and $X^4 = X^{-1}$ are each one move in the face-turn metric (FTM); rotations, slice turns, and identity moves are not in the alphabet. In particular, Σ_F is symmetric. If $w = s_1 \cdots s_m$ is a word, then $|w| = m$, and its tokens act from left to right:

$$x \cdot w = (((x \cdot s_1) \cdot s_2) \cdots) \cdot s_m.$$

This convention matters for the physical lifting below.

Let Ω denote the full 48-move FTM alphabet obtained by taking all four nonidentity powers of all 12 faces. Words over Ω may temporarily leave a subgroup; only their start and target memberships matter for an ambient FTM bound.

The part of the subgroup chain used in this work is

$$\begin{aligned} G_5 &= \langle U, R, F, L, \text{BR}, \text{BL}, \text{FR} \rangle, & G_6 &= \langle U, R, F, L, \text{BR}, \text{BL} \rangle, \\ G_7 &= \langle U, R, F, L, \text{BR} \rangle, & G_8 &= \langle U, R, F, L \rangle, \\ G_9 &= \langle U, R, F \rangle. \end{aligned}$$

For G_i we use the FTM alphabet containing the four nonidentity powers of each displayed generator of G_i ; denote it by Σ_i . Hence $\Sigma_{i+1} \subseteq \Sigma_i$ for the transitions considered here.

Definition 1 (Target-solving distance). Let $A \geq B$ be two groups in the chain, and let Γ be a symmetric FTM alphabet. For a state $x \in A$, define

$$d_\Gamma(x, B) = \min\{|w| : w \in \Gamma^*, x \cdot w \in B\},$$

and define the worst-case target-solving radius over starts in A by

$$\mathcal{R}_\Gamma(A, B) = \max_{x \in A} d_\Gamma(x, B).$$

For a phase table we use $\Gamma = \Sigma_A$ and abbreviate

$$\begin{aligned} \rho(A, B) &= \mathcal{R}_{\Sigma_A}(A, B), \\ d_{A,B}(Bx) &= d_{\Sigma_A}(x, B), \\ \mathcal{M}(A, B) &= \{Bx \in B \setminus A : d_{A,B}(Bx) = \rho(A, B)\}. \end{aligned}$$

The distance is independent of the representative of a left B -coset: if $x' = bx$ with $b \in B$, then $x' \cdot w \in B$ if and only if $x \cdot w \in B$. When $\Gamma = \Sigma_A$, $\rho(A, B)$ is the eccentricity of the target vertex B in the Schreier graph on $B \setminus A$ generated by Σ_A . It is not, by definition, the ordinary graph diameter, which maximizes distance over arbitrary pairs of vertices. We therefore use “target-solving radius” rather than “diameter” for the quantities proved by the phase tables. The combined claims below allow $\Gamma = \Sigma_5$, which may be larger than the source alphabet, and also state the resulting ambient $\Gamma = \Omega$ bounds. Their solution paths need not remain inside the source subgroup and hence should not be described as diameters of those phase Schreier graphs.

3.2 A complete physical lift of two coordinates

Let $A \geq B \geq C$. For each $p \in B \setminus A$, fix an optimal lowering word $u_p \in \Sigma_A^*$ such that

$$p \cdot u_p = B, \quad |u_p| = d_{A,B}(p).$$

The first equality implies $p = Bu_p^{-1}$. For each $q \in C \setminus B$, fix a physical representative $r_q \in B$ with $Cr_q = q$. Define the lifted state

$$\lambda(p, q) = r_q \cdot u_p^{-1}. \quad (1)$$

Thus, under left-to-right replay, the product order is the representative of the *second* coordinate followed by the inverse of the lowering word for the *first* coordinate.

Lemma 2 (Lifting and product order). *For every $(p, q) \in (B \setminus A) \times (C \setminus B)$, the state $\lambda(p, q)$ satisfies*

$$B\lambda(p, q) = p, \quad (2)$$

$$\lambda(p, q) \cdot u_p = r_q, \quad (3)$$

$$C(\lambda(p, q) \cdot u_p) = q. \quad (4)$$

Moreover, the lift is complete in the following sense. Given any $x \in A$, put $p = Bx$ and $q = C(x \cdot u_p)$. Then

$$Cx = C\lambda(p, q). \quad (5)$$

Consequently, one lifted state for each pair (p, q) covers every combined C -coset that can arise after the chosen optimal first-phase lowering word. Injectivity is neither assumed nor required.

Proof. Because $r_q \in B$ and $p = Bu_p^{-1}$,

$$B\lambda(p, q) = Br_q u_p^{-1} = Bu_p^{-1} = p.$$

Applying u_p cancels its inverse in (1), giving r_q and hence the second coordinate $Cr_q = q$.

For completeness, $p = Bx$ and $p \cdot u_p = B$ imply that $y = x \cdot u_p$ belongs to B . If $q = Cy$, then y and r_q are in the same left C -coset, so $y = cr_q$ for some $c \in C$. Therefore

$$x = y \cdot u_p^{-1} = cr_q \cdot u_p^{-1} = c\lambda(p, q),$$

which proves (5). □

The artifact materializes (1) in a full physical Megaminx state rather than combining coordinate integers. It then checks [Equations \(2\) to \(4\)](#) for every raw pair. Although the proof does not need injectivity, the two concrete lifted products in this work happen to contain no duplicate physical states.

3.3 Boundary-layer reduction

Lemma 3 (Boundary-layer reduction). *Let $A \geq B \geq C$ be finite groups with symmetric generating alphabets $\Sigma_B \subseteq \Sigma_A$. Let Γ be any symmetric FTM alphabet containing Σ_A , and put*

$$a = \rho(A, B), \quad b = \rho(B, C).$$

Construct $\lambda(p, q)$ as in [Lemma 2](#). Suppose that for every

$$(p, q) \in \mathcal{M}(A, B) \times \mathcal{M}(B, C)$$

there is a word over Γ of length at most $a + b - 1$ that takes $\lambda(p, q)$ into C . Then

$$\mathcal{R}_\Gamma(A, C) \leq a + b - 1.$$

Proof. Fix $x \in A$ and set $p = Bx$. If $d_{A,B}(p) \leq a - 1$, first apply the optimal word u_p and then solve the resulting element of B into C in at most b moves. The total is at most $a + b - 1$.

It remains to consider $p \in \mathcal{M}(A, B)$, so $|u_p| = a$. Put $y = x \cdot u_p \in B$ and $q = Cy$. If $d_{B,C}(q) \leq b - 1$, the same sequential argument costs at most $a + b - 1$. Otherwise $q \in \mathcal{M}(B, C)$. By [Lemma 2](#), $Cx = C\lambda(p, q)$. Ambient target-solving distance to C depends only on this left coset, and the assumed frontier word therefore solves x in at most $a + b - 1$ moves. These cases exhaust A . $\square$

The lemma isolates precisely the simultaneous-maximal boundary. It does not claim that a search over this boundary is complete: completeness comes from enumerating the entire product and replaying a valid bounded word for each member.

3.4 Exact phase data and exhaustive frontiers

The proof treats the implementation coordinates as group quotients only after an exact coordinate-contract audit. For each phase, the coordinate is the orbit of a tuple of labelled, oriented pieces under the source generators. The builder checks that every target generator fixes the target tuple, that the coordinate ranks and transitions are well defined, and that the BFS reaches the full declared coordinate set. A Schreier–Sims audit on the pinned 120-position actions recomputes

$$\begin{aligned} [G_5 : G_7] &= 208,099,584 \cdot 68,400, \\ [G_7 : G_9] &= 64,157,184 \cdot 25,945,920, \end{aligned}$$

as well as the two intermediate subgroup orders. To see why these checks suffice, let n_1, n_2 be the two coordinate counts in either pair. Target invariance makes each coordinate orbit a quotient by a stabilizer containing the stated target subgroup, so $n_1 \leq [A : B]$ and $n_2 \leq [B : C]$. The exact order audit gives $n_1 n_2 = [A : C] = [A : B][B : C]$; hence both inequalities are equalities and the target stabilizers are exactly B and C . Thus the coordinate fibers are the left cosets used in [Definition 1](#), rather than merely hashes that agree in sampled tests.

Proposition 4 (Phase-table data). *Complete breadth-first phase tables, regenerated from the pinned move and coordinate definitions, establish the following target-solving data.*

Phase	Transition	Source faces	$ \Sigma_A $	$ B \setminus A $	$\rho(A, B)$	$ \mathcal{M}(A, B) $
3	$G_5 \rightarrow G_6$	$U, R, F, L, \text{BR}, \text{BL}, \text{FR}$	28	208,099,584	14	212
4	$G_6 \rightarrow G_7$	$U, R, F, L, \text{BR}, \text{BL}$	24	68,400	8	2,531
5	$G_7 \rightarrow G_8$	U, R, F, L, BR	20	64,157,184	13	3,484
6	$G_8 \rightarrow G_9$	U, R, F, L	16	25,945,920	13	117

Here the radii are distances from the target cosets, in the sense of [Definition 1](#); no ordinary Schreier-graph diameter is asserted.

For phases 3 and 4, the combined quotient has

$$|G_7 \setminus G_5| = 208,099,584 \cdot 68,400 = 14,234,011,545,600, \quad (6)$$

whereas the simultaneous-maximal product contains only

$$212 \cdot 2,531 = 536,572 \quad (7)$$

raw lifts. For phases 5 and 6, the corresponding values are

$$|G_9 \setminus G_7| = 64,157,184 \cdot 25,945,920 = 1,664,617,163,489,280, \quad (8)$$

$$3,484 \cdot 117 = 407,628. \quad (9)$$

The complete physical lifting check found respectively 536,572 and 407,628 valid, pairwise distinct full states (maximum multiplicity one). The absence of duplicates makes the displayed products the exact numbers of stored obligations; the logical coverage in [Lemma 2](#) would remain valid even if duplicates had occurred.

Proposition 5 (Exhaustive certificate coverage). *The regenerated maximal-layer products admit the following exact partitions.*

Pair	Raw lifts	Exact-rewrite words	Direct words	Missing
3 + 4	536,572	203	536,369	0
5 + 6	407,628	8,461	399,167	0

Across the two pairs this is an exhaustive partition

$$944,200 = 8,664 + 935,536$$

into exact-rewrite and direct-certificate obligations. For pair 3 + 4, every replayed word has length at most 21 and in fact uses the restricted alphabet Σ_5 . For pair 5 + 6, every replayed word has length at most 25: the 8,461 exact-rewrite witnesses are words over Σ_5 , while all 399,167 direct words use the restricted alphabet $\Sigma_7 \subseteq \Sigma_5$. For every ID, the checker reconstructs the prescribed physical lift and requires membership in exactly one part of the partition. Exact-rewrite words are replayed in the full physical representation all the way to the solved state. Direct words are replayed independently in the combined quotient and in the full physical representation to the appropriate target subgroup. Thus the proposition depends on the words and their exhaustive ID coverage, not on the method that discovered them.

Remark 6 (Alphabet scope). The distinction between Σ_7 and Σ_5 is substantive. The neural database alone covers only the complement of the exact-rewrite set, so its Σ_7 words do not establish a restricted-alphabet bound for every frontier state. Pair 5 + 6 preprocessing uses the first seven faces, so the complete partition establishes a Σ_5 bound, but not a Σ_7 bound. The ambient conclusion follows because $\Sigma_7 \subseteq \Sigma_5 \subseteq \Omega$. Pair 3 + 4 likewise has a Σ_5 conclusion because both parts of its partition use that alphabet.

3.5 Certified upper bounds

Theorem 7 (Phases 3 and 4). *The combined target-solving radii satisfy*

$$\mathcal{R}_{\Sigma_5}(G_5, G_7) \leq 21, \quad \mathcal{R}_{\Omega}(G_5, G_7) \leq 21.$$

Proof. By [Proposition 4](#), the consecutive radii are 14 and 8. The simultaneous-maximal product is exhausted by [Equation \(7\)](#), and every member has a word of length at most $21 = 14 + 8 - 1$ by [Proposition 5](#). Apply [Lemma 3](#) with $(A, B, C) = (G_5, G_6, G_7)$ and $\Gamma = \Sigma_5$. The ambient statement follows because $\Sigma_5 \subseteq \Omega$. $\square$

Theorem 8 (Phases 5 and 6). *The combined target-solving radii satisfy*

$$\mathcal{R}_{\Sigma_5}(G_7, G_9) \leq 25, \quad \mathcal{R}_{\Omega}(G_7, G_9) \leq 25.$$

Proof. The two consecutive radii are 13 and 13 by [Proposition 4](#). The product in [Equation \(9\)](#) is exhaustive, and [Proposition 5](#) supplies a word of length at most $25 = 13 + 13 - 1$ for every member. The claim follows from [Lemma 3](#) with $(A, B, C) = (G_7, G_8, G_9)$ and $\Gamma = \Sigma_5$. The ambient statement follows because $\Sigma_5 \subseteq \Omega$. $\square$

Theorems [7](#) and [8](#) are upper bounds only. We do not provide lower-bound witnesses at distances 21 or 25, and therefore do not claim exact combined radii or ordinary graph diameters.

Corollary 9 (Conditional 112-move bound). *Assume that the externally reported 114-move Megaminx phase chain is a valid FTM upper bound under the same move convention, subgroup chain, and phase composition semantics used here, and that it contains the separate contributions $14 + 8$ for phases 3 and 4 and $13 + 13$ for phases 5 and 6. Then every Megaminx state is solvable in at most 112 FTM moves.*

Proof. Replacing $14 + 8 = 22$ by [Theorem 7](#)’s bound 21 saves one move, and replacing $13 + 13 = 26$ by [Theorem 8](#)’s bound 25 saves a second. All other contributions in the assumed chain are unchanged, so its total becomes $114 - 1 - 1 = 112$. $\square$

Corollary 10 (Conditional 111-move bound). *Under the hypotheses of [Corollary 9](#), assume additionally that the separately certified color-neutral first-phase reduction is compatible with the same 114-move chain and saves one further move. Then every Megaminx state is solvable in at most 111 FTM moves.*

Proof. Apply the additional one-move saving to the conditional bound in [Corollary 9](#). $\square$

The two local theorems are certified by this artifact. The corollaries are deliberately conditional because this work neither regenerates the complete external 114-move computation nor proves compatibility of every surrounding phase transition.

4 Exhaustive frontier construction

The phase tables store an exact lowering distance and one deterministic lowering move for every reachable coordinate. A maximal record additionally contains a physical 108-byte Megaminx state, a verified optimal lowering word, and masks of all possible first and last moves on optimal paths. The table builder derives the maximal layers from the generated distances; the published counts are comparison controls and are not used to manufacture or truncate those layers.

For each pair, the first maximal record is combined with the second maximal record using the lifting rule of [Lemma 2](#). The checker then verifies, for every raw pair, that

1. its first coordinate is the requested phase- i antipode;
2. the stored optimal phase- i word reaches the exact chosen phase- $(i + 1)$ representative;
3. that intermediate state has the requested second coordinate; and
4. the stored second-phase word reaches the next subgroup.

Coordinate integers are never treated as physical states. The raw order is first-layer major and second-layer minor, and an explicit raw-to-unique map is constructed without assuming injectivity. In the regenerated data all 536,572 and 407,628 physical states are distinct.

4.1 Exact preprocessing

Before neural search, the concatenation of the two optimal lowering words is processed by exact local transformations. Consecutive moves on one face are combined in the face-turn

Table 1: Exhaustive partition of the simultaneous-maximal frontiers.

Pair	Raw frontier	Exact rewrites	Direct words
3+4	536,572	203	536,369
5+6	407,628	8,461	399,167
Total	944,200	8,664	935,536

Table 2: Combined quotient and model specifications.

Property	Pair 3+4	Pair 5+6
Transition	$G_5 \rightarrow G_7$	$G_7 \rightarrow G_9$
Combined quotient size	14,234,011,545,600	1,664,617,163,489,280
State positions	120	120
Position classes	43	67
FTM actions	28	20
Random-walk $K_{\max}$	22	26
Trainable parameters	619,996	802,260
Certificate length limit	21	25

metric; commuting and cancellable moves are normalized; and meet-in-the-middle dictionaries test windows straddling the phase boundary. Every hash match is confirmed by comparing the complete physical state, and every resulting word is replayed from the corresponding frontier state.

The published reductions search through $4 + 4$ boundary windows. Both runs use the first seven faces, namely the 28 FTM moves in Σ_5 ; pair 5+6 may therefore leave its source subgroup G_7 during preprocessing. They close 203 and 8,461 states respectively. An exhaustive symmetry analysis separately reconstructs all 60 orientation-preserving rotations and retains only maps preserving all nested subgroups of the pair. In both cases the admissible stabilizer is the identity. Inversion is also rejected: it exchanges the left- and right-coset conventions, and the exhaustive frontier is not closed under it. Consequently no nontrivial orbit reduction is hidden in the coverage counts.

The remaining obligations are therefore the exact ordered ID sets shown in [Table 1](#). The equality in each row is a checked partition, not a statistical coverage estimate.

5 Neural-guided certificate generation

The direct certificates were discovered by two separately trained models, one for each combined quotient. They share an architecture and training procedure, but their state colorings, action sets, depth scales, parameters, and checkpoints are distinct. [Table 2](#) summarizes the certificate-generation configurations.

5.1 Exact quotient encodings

Each state is a coloring of the 120 oriented sticker positions induced by the target subgroup: positions movable inside the target are collapsed to one class, while the remaining positions retain fixed classes. The problem builder uses Schreier–Sims to check both the subgroup index and equality between the intended target subgroup and the pointwise stabilizer of the retained classes [20]. A simultaneous conjugacy check aligns every generated sticker action with the independent physical move convention. Thus the model sees the intended combined quotient rather than an accidental coarsening. The committed problem manifests are the

Table 3: Training configuration used for both models, except for the pair-specific random-walk limit.

Optimizer	AdamW	Learning rate	10^{-4}
Weight decay	0.003	Gradient clip	1
Batch size	1,024	Steps per epoch	256
Walk range	$2 \dots K_{\max}$	Validation size	16,384
Seed	42	Inference/training type	bfloat16

inputs to certificate replay; the artifact recomputes their group orders and physical-action alignment without loading a checkpoint.

The network first applies a position-specific class embedding. Equivalently, for coloring $x = (c_1, \dots, c_{120})$ it computes $e(x) = b + \sum_{p=1}^{120} W_{p,c_p}$ without materializing the full position-class one-hot vector. Batch normalization and the SiLU activation map this 64-dimensional representation to width 256. Each of two residual blocks has the form

$$h \leftarrow \text{SiLU}(h + \text{BN}_2(L_2(\text{Dropout}(\text{SiLU}(\text{BN}_1(L_1(h)))))$$

Dropout was zero in both published runs. A final linear layer emits one real-valued score per legal action; lower scores are searched first.

5.2 Sparse trajectory supervision

Training examples are generated online by nontrivial random walks starting at the target coordinate. Consecutive moves on the same face are excluded because their product is either one FTM move or the identity; quotient self-loops are also rejected. For a randomly selected trajectory pivot x_t , let a^- be the inverse of the preceding action and a^+ the sampled next action. Only these two outputs are supervised:

$$\mathcal{L}(x_t) = \frac{1}{2} (Q_\theta(x_t, a^-) - (t-1))^2 + \frac{1}{2} (Q_\theta(x_t, a^+) - (t+1))^2. \quad (10)$$

The diagnostic pair accuracy is the fraction for which $Q_\theta(x_t, a^-) < Q_\theta(x_t, a^+)$. The trajectory index t is not asserted to equal the shortest distance of x_t . Accordingly, Q_θ is best understood as a learned action-ranking score trained from noisy trajectory depths, not as an exact distance table or a certified Q-function.

For every sample, the walk length is uniform on the displayed integer range and the pivot is uniform among its internal vertices. Eligible nontrivial moves are sampled uniformly after the same-face exclusion. The pair-3+4 certificate run used epochs 1024 and 1120 of one training run; pair 5+6 used epochs 864 and 3744. These checkpoint identities record the run that found the certificates, not a claim that their epochs are generally optimal.

5.3 One beam per root, evaluated in batches

At each depth, every root maintains its own beam, visited set, last-face context, and paths; only neural inference is shared across roots. Scores are the current one-step values $Q_\theta(x, a)$, not cumulative path scores. At a beam of width B with A actions, the implementation first obtains the best

$$C = \min\{BA, \max(A, 16B)\}$$

parent-action pairs before CPU deduplication. It initially scans $\min\{C, \max(A, 2B)\}$ candidates and scans the rest only if duplicates prevent the next beam from filling. Same-face continuations are masked. The goal test occurs immediately after applying an action; self-loops and then previously visited keys are discarded. A key is the pair (coloring, last face),

because those two copies of one coloring have different legal continuations. The first B surviving keys in score order form the next beam.

Listing 1: One root of the batched beam search. Model calls for all active roots are fused in the implementation.

```

beam := {(root, last_face = none, word = empty)}
visited := {(root, none)}
for depth := 1 .. length_limit:
  P := legal (parent, action) pairs from beam
  rank P by one-step score Q(parent, action), ascending
  next := empty
  for pair in bounded top-candidate window P:
    child := apply(pair)
    if child is target: return pair.word
    key := (child, face(pair.action))
    if child is a self-loop or key in visited: continue
    add key to visited and pair to next
    if |next| = beam_width: break
  if next is empty: return failure
  beam := next
return failure

```

Exact floating-point ties inherit whatever ordering the PyTorch device produces; the search does not require that order to be reproducible. Any returned word is replayed in the combined quotient and on the corresponding physical state before it can enter the certificate database.

Listing 2: Survivor beam cascade used for certificate discovery.

```

U := ordered set of unresolved frontier IDs
for (checkpoint, beam_width) in schedule:
  for bounded chunk C of U:
    words := batched_per_root_beams(C, checkpoint,
                                     beam_width, length_limit)
    for (id, word) in words:
      if replay_quotient(id, word) and replay_physical(id, word):
        persist_certificate(id, word, provenance)
  U := U minus all persisted certificate IDs
  stop if U is empty
require U is empty

```

The database is resumable: existing IDs are skipped, records are tied to checksums of the physical state and proof problem, and a later stage may replace a record only with a shorter word or the same length at a smaller beam width. The production runner uses outer chunks of 10,000 frontier IDs and root minibatches that decrease from 128 at narrow widths to 16 at width 4096. The older pair-5+6 width-64 pass used an earlier low-batch implementation, which is why its throughput is not compared directly with later tiers. Beam width, checkpoint identity, and batching record discovery provenance only.

5.4 Published cascade schedules

The survivor plot in [Figure 2](#) shows the schedules; full counts, timings, and checkpoint splits appear in [Appendix A](#). Roughly 60% of each direct-certificate set was found at its first beam tier, while the maximum width was applied to only one pair-3+4 survivor and 28 pair-5+6 survivors. These observations do not isolate the causal effect of beam width. Pair 3+4 combines epoch-1024 and epoch-1120 checkpoints at width 32; pair 5+6 switches from epoch

864 at width 64 to epoch 3744 for subsequent tiers. Moreover, every later tier sees a survivor distribution selected by all earlier tiers. We therefore present the cascade as the successful certificate-generation portfolio actually used, not as evidence that one beam width dominates another.

6 Certificate verification and trusted base

The search computation answers an existential question for each frontier state: it proposes a word that appears to reach the target subgroup within the required limit. Certification turns those proposals into a finite, fail-closed proof. In particular, neither a neural score nor a claim that a beam exhausted its search space is accepted as evidence.

6.1 Certificate records and exact coverage

For each pair, the direct certificates are stored in a SQLite database. A row contains the raw state ID, a SHA-256 digest of its 108-byte physical state, the byte-coded move word and its length, and discovery provenance consisting of the beam width and checkpoint digest. A second digest binds the ID, state, word, and provenance fields. Database metadata binds the complete collection to the metric, target subgroup, proof length limit, combined-quotient definition, physical composition file, and ordered list of unresolved IDs.

Let I be the full raw frontier ID set, $E \subset I$ the IDs closed by exact rewriting, and $N \subset I$ the IDs represented in the direct-certificate database. The checker constructs I and E from regenerated artifacts and requires

$$E \cap N = \emptyset, \quad E \cup N = I. \quad (11)$$

It rejects missing IDs, extra IDs, duplicates, malformed words, stale state digests, metadata disagreement, and any word longer than the pair-specific limit. Thus a partial or interrupted search database cannot be mistaken for a complete proof.

6.2 Replay paths and the full-reproduction cross-check

The proof checker replays every accepted direct word along the first two paths below. The full-reproduction entry points additionally require the third:

1. the exact 120-position coloring of the combined quotient;
2. an independently written Python implementation of the full edge and corner permutations and orientations; and
3. a persistent Go verifier compiled against the pinned physical move definitions, receiving only the full-state bytes and canonical move word.

The quotient replay establishes arrival at the intended combined target, and the Python full-state replay applies the physical moves and tests membership in G_7 or G_9 . In a full reproduction, the Go path repeats the physical replay without parsing SQLite or importing the Python transition code. Python and Go then independently hash the canonical stream of ID, state bytes, and word; the workflow requires equality of the SHA-256 transcript digest, first and last IDs, record count, and maximum length. Thus aggregate agreement cannot hide a substituted or omitted record. Exact-rewrite witnesses are likewise replayed before their IDs enter E .

The committed historical reference reports record the complete Python replay but predate the transcript-bound Go path and contain no Go transcript fields. Accordingly, they are not evidence that this newer cross-check was executed for the published snapshot; rather, the repository makes that cross-check a fail-closed requirement of a fresh full reproduction. The

Table 4: Verification roles and their independence boundary.

Component	Implementation	Property checked
Frontier and partition	deterministic Python pipeline	complete maximal-layer product and exact ID partition
Combined replay	pair-specific Python coloring	every direct word reaches the intended combined target coordinate
Physical replay	Python <code>FullStateV1</code>	every direct word reaches the physical target subgroup
Physical cross-check	persistent Go verifier	separately implemented replay of the exact same transcript; it shares the pinned move convention
Exact rewrites	generated in Go, checked in Python	every accepted rewrite word has the allowed alphabet and length and physically solves its state

mathematical upper bounds rely on exact coverage and deterministic word replay, while the Go path is an additional implementation-diversity check on the physical action.

These checks address different failure modes. Agreement between two representations cannot by itself prove that the frontier is complete, while an exact ID partition cannot show that a stored word has the advertised action. The proof requires both [Equation \(11\)](#) and successful replay.

The deterministic coverage checker is not presented as an independent second implementation of the entire proof. Independence here is narrower: the two physical replay engines use separate code paths, and the transcript binding ensures that they receive the same direct certificates.

6.3 Regeneration checks

The full workflow begins before certificate replay. It downloads a pinned upstream source archive after verifying its digest, builds the table generator and verifiers, and regenerates the four complete phase tables. For every phase, the checker validates the reachable count, histogram, maximal distance, maximal-layer count, predecessor structure, and all stored checksums. It then reconstructs every maximal physical representative and optimal lowering word, forms the complete products from [Equations \(7\) and \(9\)](#), checks the lifting identities of [Lemma 2](#), and regenerates the exact-rewrite partition.

The published numerical controls are used only for comparison after generation. They are not loop bounds for truncating a table or manufacturing a maximal layer. Temporary outputs are written under distinct paths, and payloads with missing records or inconsistent hashes cause the run to stop.

6.4 Trusted and untrusted components

The trusted computational base consists of the pinned move and coordinate definitions, table construction, physical lifting, strict word parsers, coverage logic, cryptographic hashes as integrity guards, and the replay implementations. The neural training code, checkpoints, floating-point scores, batching order, beam widths, certificate timestamps, and GPU libraries are outside that base. The deterministic local-rewrite search is likewise a word proposer: only its checked outputs enter the proof. These components can affect which word is found and how quickly, but cannot make an invalid or absent word pass the checker. This separation is the main reason learned search is suitable here despite lacking completeness or admissibility guarantees.

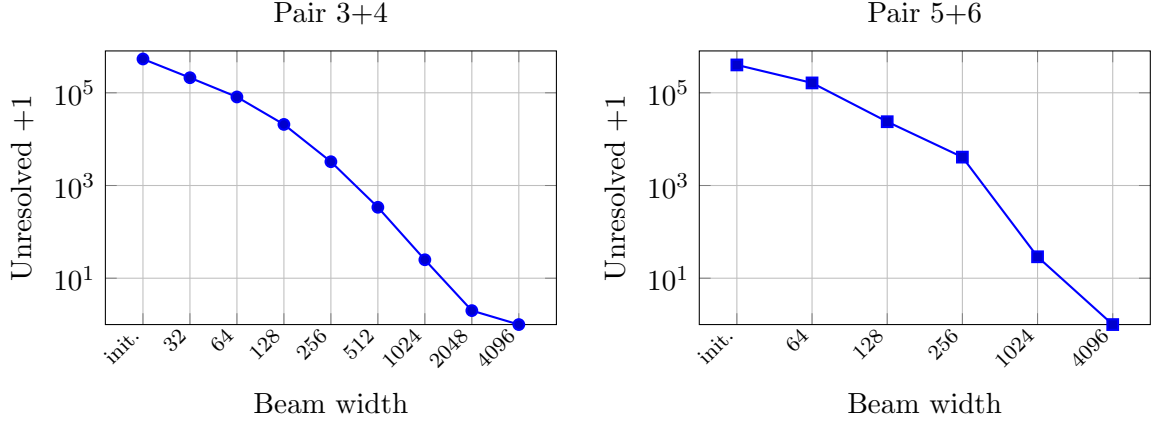


Figure 2: Unresolved direct-certificate obligations after each executed tier, plotted as count plus one so that completion is represented exactly at one. The pair-3+4 checkpoint changes within width 32; pair 5+6 changes checkpoint between widths 64 and 128.

7 Computational evaluation

We evaluate two different objects: the size reduction obtained by the boundary-layer theorem, and the observed behavior of the certificate-generation portfolio. Only the former and the final replayed coverage are proof claims. Search timings and tier-by-tier counts describe the published run and should not be read as hardware-independent bounds.

7.1 Reduction in proof obligations

The theorem replaces prohibitively large combined-quotient enumerations by the frontiers already counted in [Proposition 5](#). The reduction factors are approximately 2.65×10^7 for pair 3+4 and 4.08×10^9 for pair 5+6. Across both pairs only 944,200 physical lifts require certificates; exact preprocessing supplies 8,664 of them and learned search supplies the remaining 935,536 words.

7.2 Survivor-tail allocation

[Figure 2](#) plots the unresolved population after each executed tier. In these published runs the cascade concentrated width 4096 on the final one pair-3+4 root and 28 pair-5+6 roots, rather than running that width on the initial populations. This survivor funnel is the design rationale for the cascade and the main algorithm-engineering role of the learned component; it is not a controlled comparison with a fixed-width or non-neural solver.

On an NVIDIA A100 80GB PCIe GPU with bfloat16 inference, active model-search time was approximately 2 h 21 min 42 s for pair 3+4 and 3 h 53 min 44 s for pair 5+6, or about 6 h 15 min in total. These totals exclude phase-table construction, exact rewriting, process gaps, and final replay. Training logs through the certificate-generation checkpoints record approximately 4.6 GPU-hours in total. The point is not that every certificate had equal cost: survivors required wider tiers in this run, and those widths were applied to progressively smaller populations.

These are lengths of words found by the portfolio, not shortest-distance measurements or lower-bound witnesses. The complete histograms appear in [Appendix B](#).

Table 5: Lengths of direct neural-guided certificates. Exact-rewrite words are excluded.

Pair	Minimum	Median	Mean	Maximum	At maximum
3+4	14	20	19.78	21	172,838
5+6	14	23	22.60	25	65,139

7.3 What the cascade data do and do not establish

The published execution achieves complete coverage, but it is not a controlled ablation. Later tiers operate on distributions selected by earlier tiers, and both pairs change checkpoint during the schedule. Pair 3+4 also combines a partial epoch-1024 pass with the epoch-1120 pass at width 32, while pair 5+6 changes from epoch 864 to epoch 3744 after width 64. Consequently the rows do not identify a pure width effect, and throughput measurements from different tiers are not directly comparable.

We therefore make a narrower empirical claim: this particular portfolio found bounded, independently replayable words for all 935,536 states left after exact rewriting, within the reported hardware budget. Establishing superiority to IDA*, A*, a single fixed-width beam, or another learned policy would require a separate controlled benchmark. The certified upper bounds do not depend on the outcome of such a comparison.

7.4 Limitations

The method is tailored to a one-move improvement, where only the product of two maximal layers needs special treatment. Larger reductions would generally require certifying thicker boundary bands or a stronger structural argument. The current neural targets are noisy random-walk depths, not exact distances; the models have no admissibility guarantee. Finally, the local bounds are unconditional only for the two subgroup transitions reconstructed here. The 112- and 111-move statements inherit the external assumptions made explicit in Corollaries 9 and 10.

8 Artifact and reproducibility

The complete artifact is available at https://github.com/AnanasClassic/megaminx_34_56_phases_reduction [16]. It contains the paper sources, compressed SQLite certificate databases, both certificate-generation checkpoints for each pair, training and cascade logs, pinned problem manifests, independent verifier source, and scripts for reconstructing all omitted intermediate tables. The bibliography pins the exact artifact commit used for this manuscript; the repository landing-page URL is provided only for navigation. No DOI or third-party archival deposit is claimed at the time of writing.

The artifact supports three levels of reproduction:

1. **Package audit.** `make verify-package` streams both compressed databases, verifies compressed and uncompressed hashes, checks their schemas and complete distributions, and verifies all checkpoint hashes. It does not require a neural framework or regenerated tables.
2. **Proof reproduction.** `make verify-pair34` and `make verify-pair56` fetch the digest-pinned source snapshot, recompute the subgroup orders from the committed action manifests, rebuild the table generator and replay implementations, regenerate the phase tables, maximal records, physical products, equivalence analysis, and exact rewrites, and replay every certificate. The neural checkpoints are not loaded.

3. **Discovery reproduction.** The pair-specific training and cascade scripts retrain a model or rerun certificate discovery on a GPU. Because neural training and parallel floating-point execution need not be bit-reproducible, success is judged only by the same deterministic certificate checker.

The proof environment requires Python 3.10 or newer, SymPy for the exact Schreier–Sims audit, and the digest-pinned Go 1.22.3 container image. The published discovery run used an NVIDIA A100 80GB PCIe GPU with CUDA/bfloat16 inference. Proof replay is CPU-only. The historical problem-manifest builder also accepts a p900 generator file with a pinned SHA-256 digest; that file is not redistributed here. It is construction provenance rather than a proof-time dependency: the committed manifests are checked directly against the physical move implementation and their group orders are recomputed.

The two compressed databases contain 536,369 and 399,167 direct records. The committed reference reports are historical Python-only outputs whose schemas contain no Go transcript fields. A full current reproduction emits a new report containing the Go binary digest and the exact cross-language transcript digest. The repository also documents the 108-byte physical state format, strict FTM word grammar, coordinate conventions, artifact binary formats, and an explicit corruption-oriented threat model. Code and computational data are distributed under the repository’s stated CC BY-NC 4.0 terms, matching the license of the source dataset on which the coordinate reimplement is based.

9 Conclusion

The boundary-layer lemma turns two enormous combined subgroup-transition questions into 944,200 concrete physical obligations. Exact rewriting and neural-guided beam cascades then provide a bounded word for every obligation, while deterministic coverage and replay keep all learned components outside the proof base. The result is the unconditional pair of ambient-FTM upper bounds $\mathcal{R}_\Omega(G_5, G_7) \leq 21$ and $\mathcal{R}_\Omega(G_7, G_9) \leq 25$, and the conditional global arithmetic 112—or 111 when combined with the separate color-neutral first-phase reduction.

The broader lesson is methodological. Learned search can participate in an exhaustive computational proof without itself being complete, interpretable, or numerically trusted, provided that its output is a collection of small classical certificates, a deterministic checker proves exact coverage, and deterministic quotient and physical-state replay validates every direct word. The full-reproduction workflow additionally requires a separately implemented Go replay of the same transcript. In the published run, the survivor cascade concentrated the widest searches on the final 29 roots of a 935,536-root direct obligation set. Determining its advantage over fixed-width or non-neural alternatives requires a controlled comparison. A natural next step is to study when the same boundary-layer construction extends to thicker bands and whether controlled cascade policies can predict the appropriate search budget before running the full certificate pass.

A Published beam-cascade provenance

Tables 6 and 7 give the complete per-tier provenance retained for the certificate-generation run. Search time excludes startup, validation, database-only skips, pauses, and independent replay.

Table 6: Pair 3+4 cascade on an NVIDIA A100 80GB PCIe GPU.

Beam	Before	Newly certified	Remaining	Active time
32	536,369	324,812	211,557	1:28:13*
64	211,557	130,282	81,275	0:25:44
128	81,275	60,490	20,785	0:15:21
256	20,785	17,526	3,259	0:08:28
512	3,259	2,921	338	0:02:52
1024	338	314	24	0:00:46.9
2048	24	23	1	0:00:16.2
4096	1	1	0	0:00:00.6

*Combined epoch-1024 partial pass and epoch-1120 pass; the first interval is reconstructed from certificate timestamps.

Table 7: Pair 5+6 cascade on an NVIDIA A100 80GB PCIe GPU.

Beam	Before	Newly certified	Remaining	Active time
64	399,167	235,840	163,327	2:55:41*
128	163,327	139,544	23,783	0:40:29
256	23,783	19,689	4,094	0:10:44†
1024	4,094	4,066	28	0:06:37
4096	28	28	0	0:00:13.3

*Reconstructed from timestamps for the older low-batch implementation. †Combined across an interrupted and resumed tier.

Table 8: Exact histograms queried from the two direct-certificate databases. The exact-rewrite partition is not included.

Length	Pair 3+4	Pair 5+6	Length	Pair 3+4	Pair 5+6
14	8	1	20	168,765	32,170
15	245	17	21	172,838	54,632
16	2,631	122	22	–	72,735
17	16,279	942	23	–	79,404
18	56,097	4,392	24	–	75,502
19	119,506	14,111	25	–	65,139

B Direct-certificate length distributions

C Artifact entry points

The following commands are the intended top-level interfaces:

```
make verify-package # fast archive, schema, and checksum audit
make audit-tree # reject caches and unintended publication files
make test # unit and corruption-oriented tests
make verify-pair34 # regenerate and certify phases 3+4
make verify-pair56 # regenerate and certify phases 5+6
make paper # build this manuscript and check its LaTeX log
```

The two complete reproductions may also be invoked together with `make verify-all`. Pair-specific scripts expose training and discovery separately so that no GPU dependency is introduced into certificate replay.

D Fail-closed verification checklist

Table 9: Selected failure modes and the corresponding checks.

Failure mode	Required check
Published counts truncate generation	Derive reachability and maximal layers from regenerated tables; compare controls only afterward.
Two quotient integers are treated as a physical product	Construct a full state and round-trip both phase coordinates and the intermediate state.
Wrong multiplication or replay direction	Check the physical lifting identities and replay every stored optimal word left-to-right.
Duplicate elimination hides a raw pair	Store a raw-to-unique map and require coverage for every raw ID, irrespective of injectivity.
Search database is partial or stale	Require the exact remaining-ID set and metadata hashes of the state, problem, and reduction artifacts.
Malformed or over-length move word	Use a strict canonical parser, count FTM tokens, and enforce the pair-specific limit before replay.
Shared transition bug accepts a false word	Replay in the combined quotient, the separately implemented Python physical model, and the Go physical verifier.
Neural score is mistaken for proof	Never load a checkpoint during proof reproduction; accept only replayed words.

References

- [1] F. Agostinelli, S. McAleer, A. Shmakov, and P. Baldi. Solving the Rubik’s Cube with deep reinforcement learning and search. *Nature Machine Intelligence*, 1:356–363, 2019. <https://doi.org/10.1038/s42256-019-0070-z>.
- [2] A. Botz. Megaminx Solver v2.0. Public source-code repository, RWTH Aachen University Git-Lab. Pinned source snapshot at commit 82db40b5744297617446da888cd73d5e26f57239; <https://git.rwth-aachen.de/alexander.botz/megaminx-solver-v2.0/-/commit/82db40b5744297617446da888cd73d5e26f57239>.
- [3] A. Botz. Post 1645809 in “development of a Megaminx solver”. SpeedSolving Puzzles Community, Feb. 2025. Posted February 27, 2025; reports the merged-phase 116-move bound; <https://www.speedsolving.com/threads/development-of-a-megaminx-solver.93202/post-1645809>.
- [4] A. Botz. Post 1670603 in “development of a Megaminx solver”. SpeedSolving Puzzles Community, Sept. 2025. Posted September 22, 2025; corrects the displayed depth-14 table to depth 13 and thereby confirms, rather than improves, the previously reported 116-move total; <https://www.speedsolving.com/threads/development-of-a-megaminx-solver.93202/post-1670603>.
- [5] A. Botz. A Megaminx solver. TUdatalib software dataset, Jan. 2026. Published January 30, 2026; <https://tudatalib.ulb.tu-darmstadt.de/handle/tudatalib/5015>.
- [6] A. Botz. Post 1690628 in “development of a Megaminx solver”. SpeedSolving Puzzles Community, Mar. 2026. Identifies 114 as the improved upper bound; <https://www.speedsolving.com/threads/development-of-a-megaminx-solver.93202/post-1690628>.
- [7] A. Chervov, K. Khoruzhii, N. Bukhal, J. Naghiyev, V. Zamkovoy, I. Koltsov, L. Cheldieva, A. Sychev, A. Lenin, M. Obozov, E. Urvanov, and A. Romanov. A machine learning approach that beats large Rubik’s Cubes. arXiv:2502.13266 [cs.LG], 2025. <https://doi.org/10.48550/arXiv.2502.13266>.

- [8] E. Cohen and J. C. Beck. Heavy-tails and randomized restarting beam search in goal-oriented neural sequence decoding. In P. J. Stuckey, editor, *Integration of Constraint Programming, Artificial Intelligence, and Operations Research*, volume 12735 of *Lecture Notes in Computer Science*, pages 115–132, Cham, 2021. Springer. https://doi.org/10.1007/978-3-030-78230-6_8.
- [9] J. C. Culberson and J. Schaeffer. Pattern databases. *Computational Intelligence*, 14(3):318–334, 1998. <https://doi.org/10.1111/0824-7935.00065>.
- [10] E. Futuhi and N. R. Sturtevant. A parallel CPU–GPU framework for batching heuristic operations in depth-first heuristic search. *Proceedings of the AAAI Conference on Artificial Intelligence*, 40(43):36919–36927, 2026. <https://doi.org/10.1609/aaai.v40i43.41019>.
- [11] M. Greco, J. Toro, C. Hernández-Ulloa, and J. A. Baier. K-Focal search for slow learned heuristics (extended abstract). *Proceedings of the International Symposium on Combinatorial Search*, 15(1):279–281, 2022. <https://doi.org/10.1609/socs.v15i1.21785>.
- [12] D. Joyner. *Adventures in Group Theory: Rubik’s Cube, Merlin’s Machine, and Other Mathematical Toys*. Johns Hopkins University Press, Baltimore, 2 edition, 2008. <https://doi.org/10.56021/9780801890123>.
- [13] R. E. Korf. Finding optimal solutions to Rubik’s Cube using pattern databases. In *Proceedings of the Fourteenth National Conference on Artificial Intelligence and Ninth Conference on Innovative Applications of Artificial Intelligence*, pages 700–705. AAAI Press, 1997. <https://cdn.aaai.org/AAAI/1997/AAAI97-109.pdf>.
- [14] D. Kunkle and G. Cooperman. Twenty-six moves suffice for Rubik’s Cube. In *Proceedings of the 2007 International Symposium on Symbolic and Algebraic Computation*, pages 235–242. Association for Computing Machinery, 2007. <https://doi.org/10.1145/1277548.1277581>.
- [15] V. Kuznetsov. The exact worst-case color-neutral distance for the first block phase of the Megaminx. Reproducibility package for a forthcoming manuscript, July 2026. Version 068bf7bc78049b6046b8b63a580202af48158fbe; https://github.com/AnanasClassic/megaminx_phase1_color_neutral_distance/tree/068bf7bc78049b6046b8b63a580202af48158fbe.
- [16] V. Kuznetsov. Megaminx phase 3+4 and 5+6 reductions: Computational artifact. GitHub repository, July 2026. Version d572b412af6692b81e10b3cf27e4353c149d55a1; https://github.com/AnanasClassic/megaminx_34_56_phases_reduction/tree/d572b412af6692b81e10b3cf27e4353c149d55a1.
- [17] T. Rokicki. Post 1675762 in “development of a Megaminx solver”. SpeedSolving Puzzles Community, Nov. 2025. Posted November 6, 2025; reports 24-move solutions, after adjoining a third face generator, for all 16 distance-26 two-generator antipodes. This reduces the last-phase upper bound from 26 to 25 and the external total from 116 to 115; <https://www.speedsolving.com/threads/development-of-a-megaminx-solver.93202/post-1675762>.
- [18] T. Rokicki. Post 1675972 in “development of a Megaminx solver”. SpeedSolving Puzzles Community, Nov. 2025. Posted November 8, 2025; reports that all 7,595 distance-25 two-generator positions, reduced to 1,199 representatives by symmetry and inversion, are solvable with the enlarged generator set in at most 24 moves. This further reduces the last-phase upper bound from 25 to 24 and the external total from 115 to 114; <https://www.speedsolving.com/threads/development-of-a-megaminx-solver.93202/post-1675972>.
- [19] T. Rokicki, H. Kociemba, M. Davidson, and J. Dethridge. The diameter of the Rubik’s Cube group is twenty. *SIAM Journal on Discrete Mathematics*, 27(2):1082–1105, 2013. <https://doi.org/10.1137/120867366>.
- [20] Á. Seress. *Permutation Group Algorithms*, volume 152 of *Cambridge Tracts in Mathematics*. Cambridge University Press, Cambridge, 2003. <https://doi.org/10.1017/CB09780511546549>.
- [21] K. Takano. Self-supervision is all you need for solving Rubik’s Cube. *Transactions on Machine Learning Research*, 2023. <https://openreview.net/forum?id=bnBeNFB27b>.